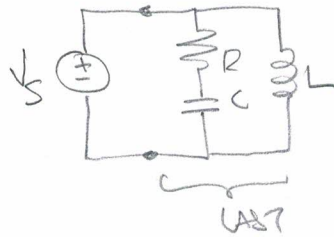


7.6)



$$\omega = 377 \text{ rad/sec}$$

$$L = 0.08 \text{ H}$$

$$R = 10 \Omega$$

$$C = 250 \text{ nF}$$

$$V_s = 50 \angle 0^\circ \text{ Volt RMS}$$

RESKIM EFFEKTIF DAN LAGUEN FURBROUKAR

LOSONG: LAGUEN IMPEDANS BUK

$$Z_{LST}^{-1} = \frac{1}{R + \frac{1}{j\omega C}} + \frac{1}{j\omega L} =$$

$$= \frac{j\omega L + R + \frac{1}{j\omega C}}{j\omega L (R + \frac{1}{j\omega C})}$$

$$\Rightarrow Z_{LST} = \frac{j\omega L (R + \frac{1}{j\omega C})}{j\omega L + R + \frac{1}{j\omega C}} \quad (1)$$

KOMPLEK EFFEKTIF DAN LAGUEN FURBROUKAR

$$S = \frac{1}{2} V I^* = V_{RMS} I_{RMS}^*$$

MEN

$$I_{RMS} = \frac{V_{RMS}}{Z} \Rightarrow S = V_{RMS} \frac{V_{RMS}^*}{Z_{LST}^*} =$$

$$= |V_{RMS}|^2 \frac{1}{Z_{LST}^*} \quad (2)$$

(1) ger

$$Z_{LST} = \frac{j\omega L (1 + j\omega C R)}{1 - \omega^2 L C + j\omega C R} =$$

$$= \frac{-\omega^2 L C R + j\omega L}{1 - \omega^2 L C + j\omega C R}$$

NUMERISIR

$$Z_{LST} = \frac{-22.74 + j30.16}{-1.27 + j0.754} = \frac{37.77 \angle 142.98^\circ}{1.48 \angle 120.70^\circ}$$

Q

$$Z_{\text{last}} = 25.52 / 22.28^\circ \Omega$$

Wirkleistung BLK (DEN RECHN)

$$P = |V_{\text{rms}}|^2 \operatorname{Re} \left(\frac{1}{Z_{\text{last}}^*} \right) = 50^2 \operatorname{Re} \left(\frac{1}{25.52 \angle -22.28^\circ} \right) =$$

$$= 50^2 \frac{1}{25.52} \cos(22.28^\circ) \approx \underline{91.73 \text{ WATT}}$$

$$\underline{S_{\text{VAR}}} = P \approx 91.7 \text{ W}$$